

Md Faiyaz Khan

Kolkata, India | +91 8777488856 | faiyazkhan.work@gmail.com | faiyazkhan.in | [LinkedIn](#) | [GitHub](#)

PROFESSIONAL SUMMARY

Aspiring Machine Learning Engineer with hands-on experience building end-to-end machine learning systems, from data auditing and feature engineering to model evaluation, explainability, API development, and deployment. Skilled in developing leakage-safe ML pipelines, explainable decision-support systems, and deployable machine learning applications using Python, Scikit-learn, FastAPI, Streamlit, Docker, and SQL. Built practical machine learning projects through structured coursework and extensive self-directed learning.

PROJECTS

UPI Fraud Detector | *Python, Scikit-learn, Random Forest, FastAPI, Streamlit, Docker, SHAP* [GitHub](#)

- Developed an end-to-end fraud detection and risk scoring system for UPI transactions using machine learning, explainability, and real-time prediction workflows.
- Performed extensive dataset auditing and multi-stage data leakage detection, identifying and removing 34 leakage features through statistical analysis and model diagnostics.
- Engineered leakage-safe preprocessing pipelines with train-only frequency encoding, feature scaling, and consistent train-serving transformations.
- Evaluated Logistic Regression, Random Forest, and XGBoost using ROC-AUC, PR-AUC, and stratified cross-validation; selected Random Forest achieving 0.9967 test ROC-AUC and 0.9867 PR-AUC on the cleaned feature set.
- Built an explainable fraud decision-support system using SHAP, FastAPI, Streamlit, Docker, GitHub Actions CI, and automated testing with Pytest.

SureVisit | *Python, Scikit-learn, Random Forest, FastAPI, Streamlit, Docker, SHAP* [GitHub](#)

- Developed a healthcare appointment no-show prediction and decision-support system to identify high-risk patients and recommend operational interventions.
- Engineered temporal and historical features including leakage-safe attendance history and patient risk profiling using chronologically ordered expanding-window computations.
- Trained and evaluated Logistic Regression, Decision Tree, Random Forest, and XGBoost models; selected Random Forest achieving 0.7425 cross-validated ROC-AUC and 0.80 recall for missed appointments.
- Integrated SHAP explainability to provide interpretable patient-level predictions and actionable risk recommendations.
- Designed and deployed a production-style architecture using FastAPI, Streamlit, Docker, Pytest, and reusable train-serving feature engineering pipelines.

TECHNICAL SKILLS

Languages: Python, SQL

Machine Learning & Data Science: Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn

Backend & Databases: FastAPI, MySQL, PostgreSQL, Microsoft SQL Server

Deployment & Tools: Streamlit, Docker, Git, GitHub, GitHub Actions

EDUCATION

Intellipaath (in collaboration with IIT Patna)

2025 – 2026

Minor Program in Generative AI and Machine Learning

Coursework and hands-on projects

- Completed selected coursework and practical projects covering machine learning fundamentals, feature engineering, model evaluation, machine learning pipelines, explainable AI, and deployment workflows through structured learning and independent project development.